ReLU for the linear model

1) Model equation:

$$y = f[x, \varphi] = \varphi_0 + \varphi_1 \cdot a[\overbrace{\theta_{10} + \theta_{11} x}^{z}] \quad \Big\} \quad \text{4 parameters: } \varphi_0, \varphi_1, \theta_{10}, \theta_{11}$$

2) No activation: $a[z] = z \ \forall z$

$$y = \varphi_0 + \varphi_1 (\theta_{10} + \theta_{11} x) = \underbrace{(\varphi_0 + \varphi_1 \theta_{10})}_{\text{intercept}} + \underbrace{\varphi_1 \theta_{11}}_{\text{slope}} x = f_*$$

$\longrightarrow$ a straight line

3) ReLU

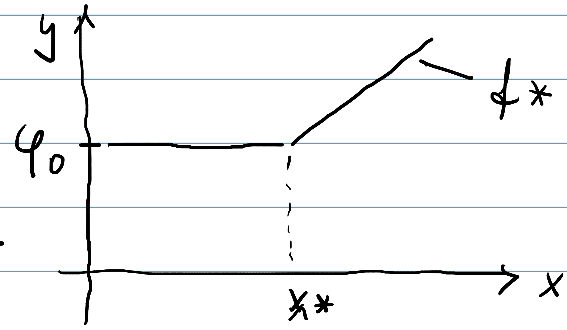
$$a[z] = \begin{cases} 0 & \forall z < 0 \\ z & \forall z \geq 0 \end{cases}$$

z : pre-activation function

$z = (\theta_{10} + \theta_{11} x)$

4) Where's the kink?
(Where does the unit switch on/off?)

$$z = 0 \Rightarrow \theta_{10} + \theta_{11} x = 0 \Rightarrow x^* = -\frac{\theta_{10}}{\theta_{11}}$$



5) "Off" branch:

$$\forall x : \theta_{10} + \theta_{11} \cdot x < 0 : a[\cdot] = 0 \Rightarrow y = \varphi_0 \begin{cases} \text{flat line} \\ \text{zero slope} \end{cases}$$

6) "On" branch:

$$\forall x \ \theta_{10} + \theta_{11} \cdot x \geq 0 : a[\cdot] = \mathbb{1}$$

identity = passes input through unchanged: $f(z) = z \cdot \mathbb{1}$

$$\Rightarrow y = \varphi_0 + \varphi_1 (\theta_{10} + \theta_{11} x) = (\varphi_0 + \varphi_1 \theta_{10}) + (\varphi_1 \theta_{11}) x$$

7) Numeric example:

$\varphi_0 = 0,\ \varphi_1 = 2,\ \theta_{10} = -1,\ \theta_{11} = 1 \Rightarrow -\frac{\theta_{10}}{\theta_{11}} = \underbrace{x^*}_{\text{kink}} = 1$

x	-1	0	0.5	1	1.5	2	3
y	0	0	0	0	1	2	4

$$\boxed{y = -2 + 2x}$$

8) Code check

```
import numpy as np
import matplotlib.pyplot as plt

def relu(z):
    return np.maximum(0, z)

def y(x, phi0, phi1, theta10, theta11):

    return phi0 + phi1 * relu(theta10 + theta11 * x)

x = linspace(-2, 4, 100)
plt.plot(x, y(x, phi0=0, phi1=2, theta10=-1, theta11=1))

plt.axvline(1, linestyle='--', color='gray')
plt.xlabel('x'); plt.ylabel('y')
plt.show()
```